

# E-Voting Through Blockchain

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# Abstract:

**A Blockchain-Based Voting System ensures secure, transparent and tamper-proof elections using Ethereum smart contracts. It allows voters to register, verify identity via OTP, and cast votes anonymously using a decentralized system.**

# Objectives:

1

Decentralized Voting

2

Security and Transparency

3

Voter Authentication

4

Real-Time Monitoring

5

Immutable ledger

6

Accessibility and  
Cost-Efficiency

| Feature                       | <u>Existing Voting Systems</u>                               | <u>Blockchain-Based Voting System</u>                           |
|-------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------|
| • <b>Centralization</b>       | Controlled by a central authority                            | Fully decentralized using blockchain                            |
| • <b>Security</b>             | Vulnerable to hacking, tampering, and errors                 | Cryptographically secure and tamper-proof using smart contracts |
| • <b>Transparency</b>         | Limited visibility, controlled by election authorities       | Fully transparent with votes recorded on an immutable ledger    |
| • <b>Result Accuracy</b>      | Dependent on manual counting and EVM reliability             | 100% accurate as results are executed via smart contracts       |
| • <b>Voter Authentication</b> | Manual verification, slow and error-prone                    | Aadhaar-based OTP verification for better security              |
| • <b>Vote Manipulation</b>    | Susceptible to rigging, miscounting                          | Votes are immutable and cannot be altered once recorded         |
| <b>Real-Time Monitoring</b>   | Limited tracking with delayed results                        | Admins/users can monitor votes and election phases in real time |
| • <b>Tamper Resistance</b>    | Data can be altered by insiders or hackers                   | Immutable ledger prevents vote modifications                    |
| • <b>Cost Efficiency</b>      | Expensive due to manpower, paper ballots, and security costs | Reduces costs by automating and removing intermediaries         |

# *Problem Statement:*

**Traditional Voting Systems face Challenges like vote manipulation, lack of transparency and accessibility issues. This System addresses these problems by leveraging Blockchain Technology to create a trustworthy, verifiable and decentralized voting process.**

# Proposed Methodology And Implementation:

The proposed Blockchain-Based Voting System ensures a secure, transparent, and decentralized voting process by leveraging Ethereum smart contracts and Web3.js for blockchain interactions. The methodology involves the following key aspect:

The System is divided into three -Layers:

## **a)User Interface Layer(Frontend)**

- Built using **React.js** to provide an interactive web-Based platform.
- Users(Voters,Election Officials and Admins) interact with the System through the **web Application**.
- **MetaMask** integration allows voters to sign **blockchain transactions** securely.

## b)Blockchain & Smart Contract Layer :

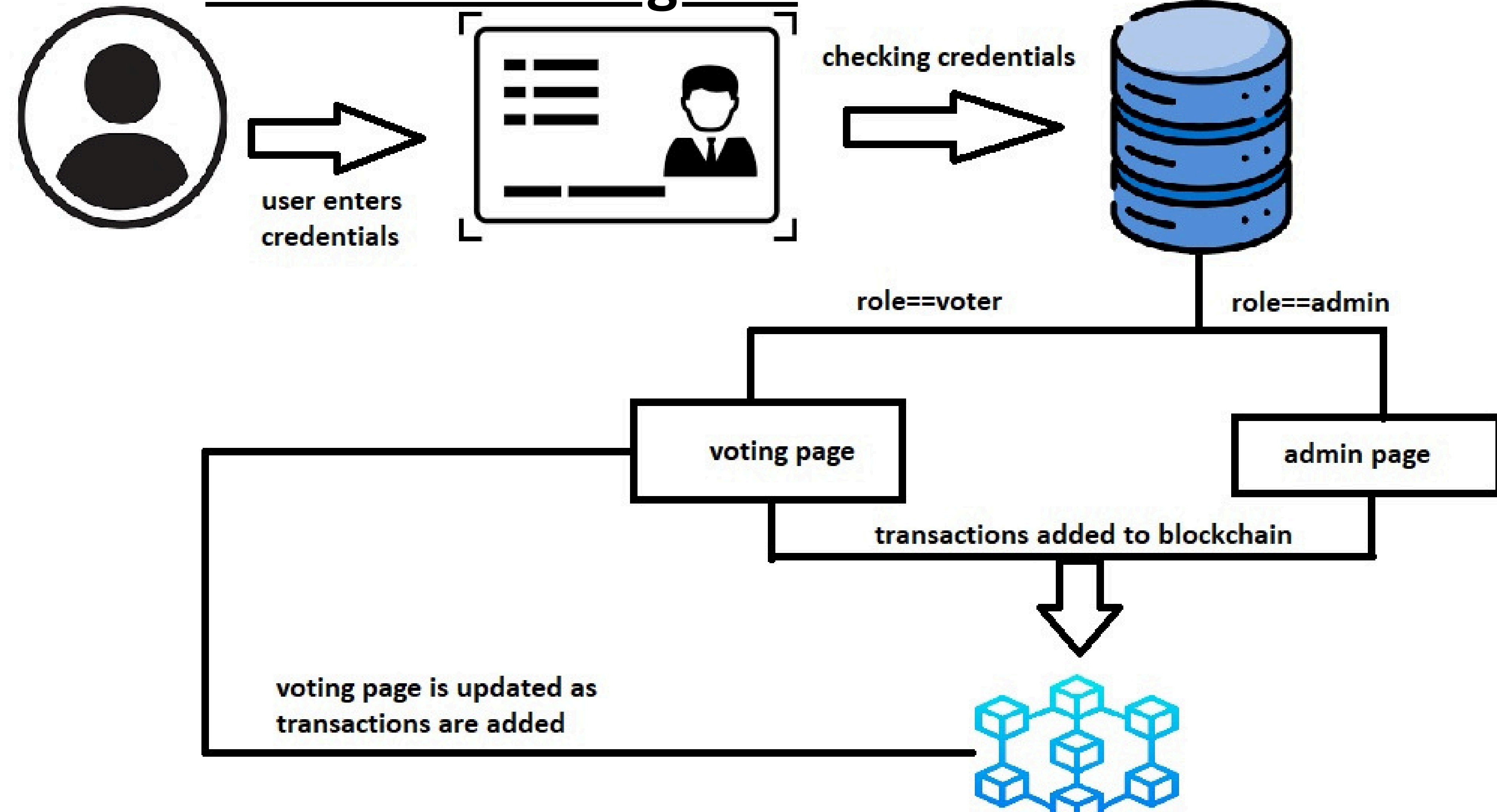
**Ethereum smart contracts**(written in Solidity) handle:

- Voter registration and verification
- Secure vote casting
- Automatic vote tallying and result declaration.

## c)Backend Layer:

- Developed using **Express.js** to handle server-side logic
- **OTP-Based Aadhar verification** ensures only valid voters can participate
- Uses **Web3.js** to interact with the Ethereum blockchain.
- **Ganache/sopolia** network is used for local blockchain deployment and testing.

# Architecture Diagram:

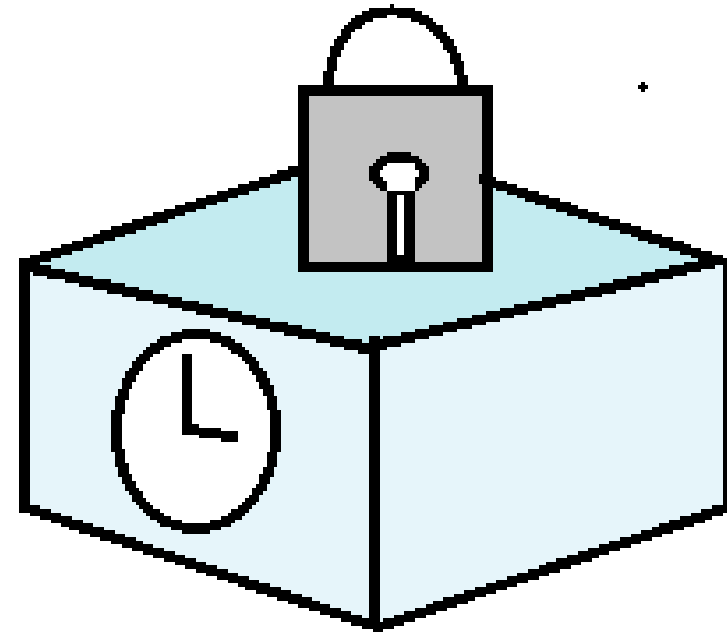


## Election Phases:

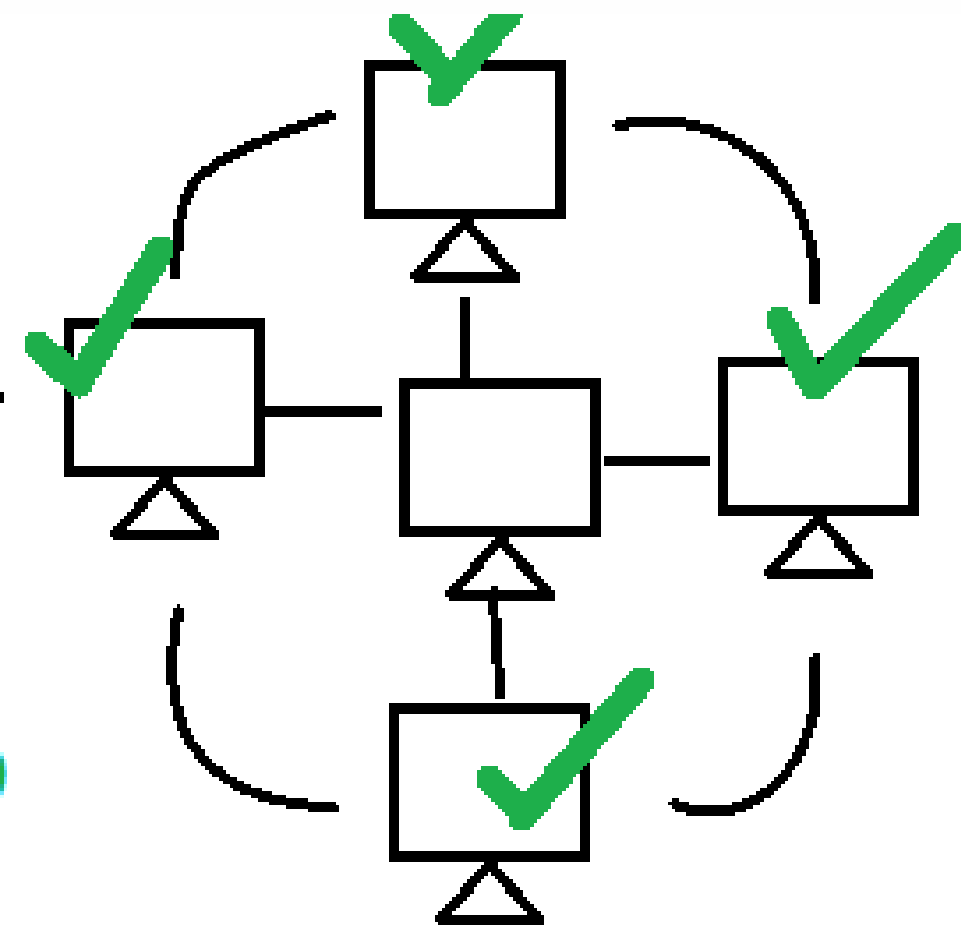
- **Voter Registration:** Users register with Aadhaar number, name, and *MetaMask wallet address*. OTP authentication verifies identity.
- **Voter Verification:** Admin manually verifies registered voters to prevent duplicates.
- **Voting Process:** Verified users securely cast votes via blockchain. Each vote is recorded as an Ethereum smart contract transaction.
- **Result Declaration:** System fetches real-time vote counts from the blockchain. Results are displayed transparently.

Name  
Password  
☐ A  
☒ B  
☐ C  
VOTE

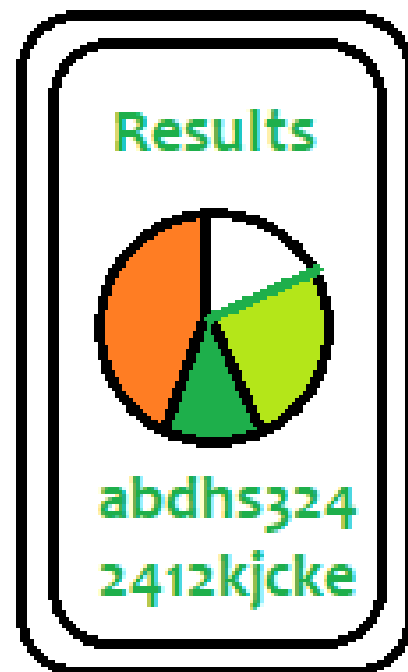
Voter UI



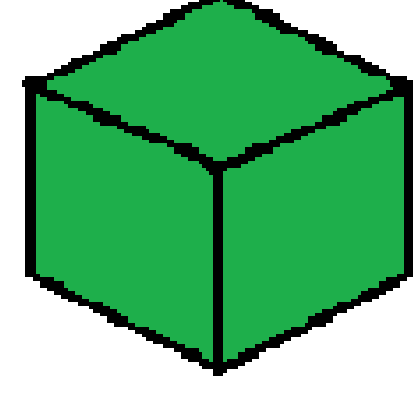
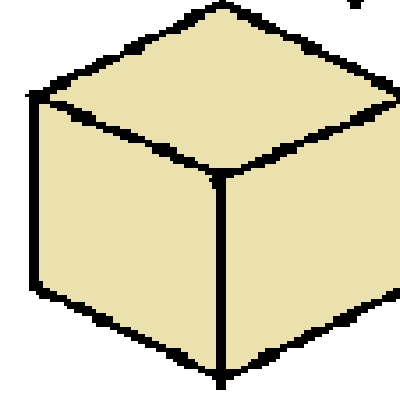
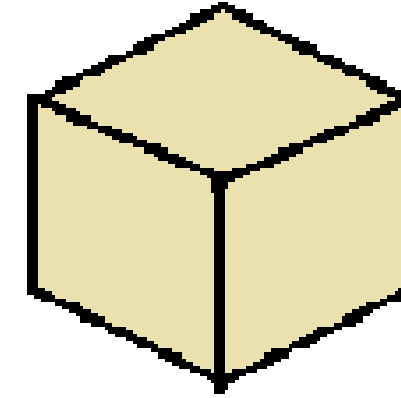
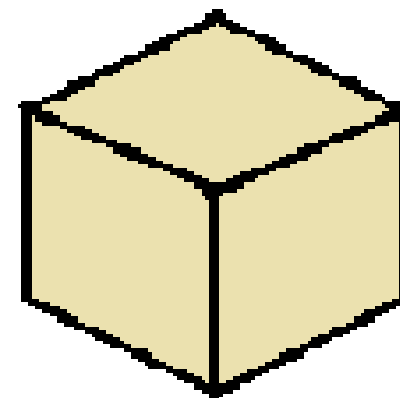
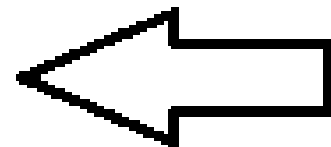
TransactionID, Timestamp ( abdhs3242412kjckelz..)



All details are broadcasted to the network, where each node verifies it.



Voter can view results soon after voting and can trace back.



Transaction (Vote) is added into the chain

# Results:

1

Transparency and Security

2

Efficiency and Speed

3

Elimination of Fraud

4

Voter Privacy

5

Scalability and Cost Reduction

6

Trust and Adoption

# Conclusion:

- Ensures a fair, secure, and transparent voting system
- Eliminates fraud and manipulation through blockchain technology
- Provides instant and tamper-proof election results
- Enhances voter privacy and trust in the electoral process
- Scalable for large-scale elections with reduced operational costs
- Encourages adoption in government and private sector elections



Thank you!